experience, and randomising the simulator across a family of plants produces policies that transfer without being tuned to one machine. The second is that the actuator itself can be learned. Rather than deriving valve and cylinder behaviour from first principles and fitting its parameters, a network trained on recorded motion can represent the command-to-motion relationship directly, including the nonlinearities that analytical models approximate away

Combining the two is attractive and not straightforward. A learned actuator model must be cheap enough to evaluate inside the training loop, which rules out the high-fidelity models used for offline analysis, and it must remain valid across the family of machines the policy will meet. This has motivated a line of work in which a compact code describing the current plant is inferred from recent interaction history and supplied to the controller so that a single policy can serve a family rather than a single machine. Such methods report substantial gains in legged locomotion and in manipulation, and they are the direct antecedent of the architecture examined here.

1.3 Challenges of Precise Tracking with Hydraulic Mowing Arms

A valve command does not produce a proportional cylinder motion. The relationship is nonlinear, depends on load and supply pressure, and is subject to dead time between the command and any measurable response. It also differs between machines: seal condition fluid temperature, hose compliance and the attachment fitted all change it, so two machines of the same model respond differently and one machine responds differently across a season. A controller that inverts only the geometry therefore leaves the largest source of tracking error unaddressed.

That difficulty admits two responses, and they are routinely conflated. The first is robustness: train across a randomised family so that a single fixed model is adequate everywhere. It costs nothing at run time and gives up the difference between the family and the individual. The second is adaptation: infer which member of the family is present from the interaction history and condition the model or the policy on that estimate. It promises to recover that difference, at the cost of an estimator that must work from observable signals alone, since a deployed machine has no access to its own plant parameters.

The second response is only worth pursuing under conditions that are rarely checked before the work is done. Plant identity must be recoverable from the data the machine actually produces, which depends on how the machine is being driven while that data is collected. And the difference between knowing the plant and not knowing it must be large enough to matter once the estimate reaches a feedback controller. Neither condition is guaranteed, both are specific to the task, and both can be measured before an estimator is built. Establishing those measurements, and applying them to a hydraulic mowing arm is the problem this thesis addresses.

1.4 Contribution of This Thesis

The difficulty described in Section 1.3 is a modelling problem before it is a control problem. A hydraulic actuator can be simulated accurately, but the models that do so are too slow to sit inside a reinforcement-learning loop that requires millions of environment steps